# Study of Low Temperature Cu-to-Cu Bonding using Reducing Plasma

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***Abstract*— This study investigates the effect of $CH_4$ plasma treatment on the quality of Cu-to-Cu bonding. An 8-inch notch-type Si wafer with a 700 nm $SiO_2$ layer is utilized. A Ti adhesion layer of 50 nm and a 1 µm Cu thin film are subsequently deposited onto the $Si/SiO_2$ wafer using DC magnetron sputtering. Following the Ti/Cu deposition, $CH_4$ plasma treatment is applied using inductively coupled plasma chemical vapor deposition (ICP-CVD). The influence of $CH_4$ on Cu surface is optimized with power that affect plasma concentration, and gas flow ratio (Ar:$CH_4$) that can directly affect reaction mechanisms or surface changes. Cu-to-Cu bonding is performed at 260°C for 1 hour, followed by annealing at 200°C for 1 hour. Oxides and contaminants on the Cu surface are removed and the surface roughness decreases as plasma-treatment time increases, as confirmed by X-ray photoelectron spectroscopy(XPS) analysis and atomic force microscopy(AFM) measurement. Bonding quality is evaluated using SAT, FE-SEM, TEM, which shows that the plasma-treated specimen has fewer voids at the bonding interface and more Cu diffusion. The results show that $CH_4$ plasma treatment enhances Cu-toCu bonding quality.**

***Keywords*— Hybrid Cu bonding, Reducing Plasma treatment, Reduction of Cu oxide, Plasma pretreatment***

## I. Introduction

High Bandwidth Memory (HBM) is essential for High-Performance Computing (HPC) and AI accelerators. As the number of I/Os increase exponentially, reducing the size of bumps for HBM interconnects has become a critical focus. However, achieving sub-micron pitch sizes (<10 µm) is challenging with conventional solder bump technology [1]. Hybrid bonding technology offers a solution by enabling sub-micron pitch interconnects. One of the key challenges in chip-to-chip hybrid bonding is the removal of copper oxide and the prevention of further oxidation, both of which are crucial for lowering bonding temperatures and ensuring chip reliability during the process [2].

Various plasma pre-treatment methods, such as Ar, $N_2$, $O_2$ plasma, have been studied for their ability to remove oxides and protect the Cu surface [3-4]. However, These plasma treatments primarily rely on physical removal mechanisms, including surface bombardment, which can result in surface damage and undesired etching of the bonding surface [5]. Additionally, forming gas($Ar/H_2$) plasma treatment was also studied as a Cu reducing agent, utilizing both chemical and physical mechanisms to reduce Cu oxide [6]. While H radicals and H ions are effective at facilitating this reduction, the low hydrogen concentration in forming gas(typically 95% Ar and 5% $H_2$) limits its ability to achieve complete oxide removal.

Hence, an alternative plasma treatment capable of generating a higher concentration of hydrogen radicals and ions is essential. $CH_4$ plasma can be identified as a promising candidate to fulfill this requirement. Compared to hydrogen plasma, $CH_4$ plasma exhibits significantly superior metal reduction efficiency according to the following reaction [7] :

$$Cu_2O + H_2 \rightarrow H_2O + 2Cu \quad (1)$$

$$4Cu_2O + CH_4 \rightarrow CO_2 + 2H_2O + 8Cu \quad (2)$$

$$3Cu_2O + CH_4 \rightarrow CO + 2H_2O + 6Cu \quad (3)$$

Furthermore, $CH_4$ plasma treatment is anticipated to form hydride carbonaceous layers on the surface, which can inhibit re-oxidation and enhance bonding efficiency. These layers, formed by hydrogen and carbon atoms, do not interfere with the bonding process due to their small atomic radii, as they diffuse into Cu during bonding and dissipate without leaving a significant impact [8].

## II. Methods

### *2.1 Experimental*

The experiment utilized an 8-inch notch-type Si wafer coated with a 700 nm $SiO_2$ layer. To enhance adhesion, a 50 nm Ti layer was deposited, followed by the deposition of a 1 µm Cu thin film using DC magnetron sputtering. The sputtering process was carried out under a working pressure of 5 mTorr with a DC power of 2000 W and an Ar gas flow rate of 100 sccm. After completing the Ti/Cu deposition, $CH_4$ plasma treatment was performed using an inductively coupled plasma chemical vapor deposition (ICP-CVD). The process variables were power and gas flow rate, and pressure and processing time were fixed. The process parameters for $CH_4$ plasma optimization are summarized in Table 1. Cu-to-Cu bonding was conducted at 260°C for 1 hour, followed by a subsequent annealing step at 200°C for an hour. The sample preparation and process flow are illustrated in Figure 1.

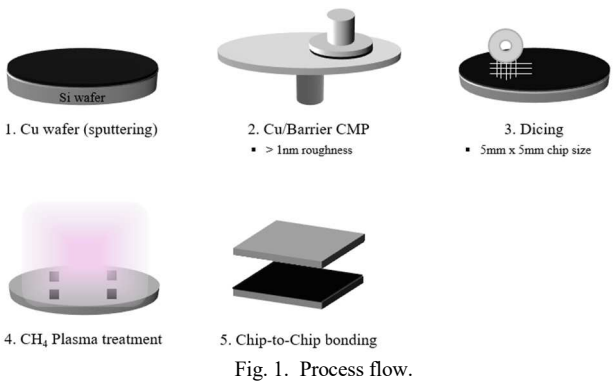



Fig. 1. Process flow.

TABLE I. PROCESS PARAMETERS

| MAIN EFFECT | PARAMETERS | RANGE |
|---|---|---|
| PLASMA CHARACTERISTICS | ICP POWER | [3OOW-1200W] |
| SURFACE REACTION | AR:$CH_4$ GAS RATIO | [1:1 – 1:3] |

*2.2 Analysis*

The Cu surface's chemical state is examined using X-ray photoelectron spectroscopy (XPS), and atomic force microscopy (AFM) is employed to measure surface roughness. Hydrophilicity is evaluated through water contact angle (WCA) measurements. The quality of Cu bonding is verified using scanning acoustic tomography (SAT), with additional structural characterization planned through scanning electron microscopy (SEM) and transmission electron microscopy (TEM). These are summarized in Table 2.

TABLE II. EVALUATION METHODS

| PURPOSE | ANALYSIS APPROACH |
|---|---|
| SURFACE ANAYLSIS | XPS, AFM, WCA |
| BONDING ANAYLSIS | SAT, SEM, TEM |

## III. RESULTS AND DISCUSSIONS

XPS analysis reveals a shift in the Cu 2p binding energy to lower values, indicating the occurrence of a reductive phenomenon. Additionally, a minor formation of a C-H layer was observed, suggesting the potential development of a passivation layer on the Cu surface. Surface roughness ($R_q$), measured via AFM, exhibited a decrease with $CH_4$ plasma treatment on the Cu surface.

Cu-Cu bonding was carried out at 260ºC using die-to-die bonding process, and post-bonding evaluation using SAT confirmed that the bonding interface was improved in the $CH_4$ plasma treated sample compared to the non-plasma treated sample.

$CH_4$ plasma treatment provided a reduction process of Cu oxide to Cu and showed the formation of a $CH_x$ passivation layer that prevented oxidation of the Cu surface. The $CH_4$ chemical reaction mechanisms will be further discussed.

## IV. SUMMARY

This study investigates the use of $CH_4$ plasma for Cu surface reduction. $CH_4$ plasma treatment prior to Cu-to-Cu bonding serves two primary functions. First, it removes surface impurities and reduces $Cu_2O$ (cuprous oxide) present on the surface. Second, the hydride carbonaceous layer formed during $CH_4$ plasma treatment prevents reoxidation during the die-to-die bonding process. Consequently, $CH_4$ plasma pretreatment of the Cu surface enables bonding at lower temperatures, enhancing both the bonding quality. These attributes are critical for advanced 3D stacking applications.


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